

Math 241, F04
Quiz 9

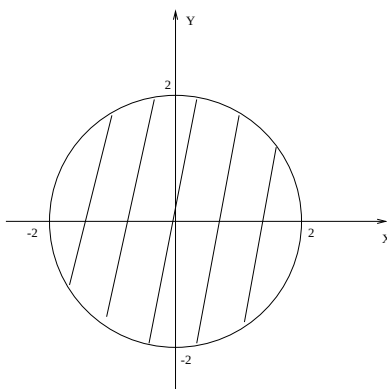
Name:
SSN: xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (10 points)

a) Let S be the region enclosed by $x^2 + y^2 = 4$, sketch it and re-write it down in the form of polar rectangle.

Solution:



$$S = \{(r, \theta) \mid 0 \leq r \leq 2, 0 \leq \theta \leq 2\pi\}$$

b) Evaluate $\iint_S e^{x^2+y^2} dA$ by using polar coordinates.

Solution:

$$\begin{aligned} \iint_S e^{x^2+y^2} dA &= \int_0^{2\pi} \int_0^2 e^{r^2} r dr d\theta \\ &= \int_0^{2\pi} \left[\frac{e^{r^2}}{2} \right]_0^2 d\theta \\ &= \int_0^{2\pi} \frac{1}{2} (e^4 - 1) d\theta \\ &= 2\pi \cdot \frac{1}{2} (e^4 - 1) \\ &= \pi(e^4 - 1) \end{aligned}$$

Problem 2. (10 points) Find the mass m and center of mass $(\bar{x}, \bar{y})$ of the lamina bounded by the lines $x = 0, x = 4, y = 0, y = 3$ with the density $\delta(x, y) = y + 1$.

Solution: It is clear that the lamina can be represented by

$$S = \{(x, y) \mid 0 \leq x \leq 4, 0 \leq y \leq 3\}$$

Then,

$$\begin{aligned} m &= \iint_S \delta(x, y) \, dA = \int_0^4 \int_0^3 (y + 1) \, dy \, dx \\ &= \int_0^4 \left[\frac{y^2}{2} + y \right]_0^3 \, dx = \int_0^4 \frac{15}{2} \, dx = 4 \cdot \frac{15}{2} = 30 \end{aligned}$$

$$\begin{aligned} M_y &= \iint_S x \delta(x, y) \, dA = \int_0^4 \int_0^3 x(y + 1) \, dy \, dx \\ &= \int_0^4 x \left[\frac{y^2}{2} + y \right]_0^3 \, dx = \int_0^4 \frac{15}{2} x \, dx = \left[\frac{15x^2}{4} \right]_0^4 = 60 \end{aligned}$$

$$\begin{aligned} M_x &= \iint_S y \delta(x, y) \, dA = \int_0^4 \int_0^3 (y^2 + y) \, dy \, dx \\ &= \int_0^4 \left[\frac{y^3}{3} + \frac{y^2}{2} \right]_0^3 \, dx = \int_0^4 \frac{27}{2} \, dx = 4 \cdot \frac{27}{2} = 54 \end{aligned}$$

So we have

$$\bar{x} = \frac{60}{30} = 2, \quad \bar{y} = \frac{54}{30} = \frac{9}{5}.$$